

PA4 Writeup

TODO 2.2.1 - Best Performing Prompting Strategies

The most effective prompts were those that described highly distinctive, visually unambiguous color combinations unique to a single class. For example, the Green Jay prompt ("a brightly colored bird with vivid green wings, a blue head, and yellow underparts") performs well because no other class in the dataset shares that three-color combination. CLIP's visual-language alignment encodes color as a strong signal, so prompts anchored in specific hue patterns give the model a clear separating boundary.

Similarly, the Arctic Tern prompt ("a white seabird with a black cap, bright red bill, deeply forked tail, and long wings") succeeds because it stacks multiple distinctive features: habitat (seabird), coloring (white/black), and morphology (forked tail, red bill). The conjunction of these features leaves little ambiguity. The White-necked Raven prompt also performs reliably, as an otherwise all-black bird with a single white patch is a distinctive pattern that CLIP maps cleanly to images containing that contrast.

The general principle is: specificity of visual contrast beats descriptive completeness. A prompt that identifies one or two features that are unique to that class outperforms a longer prompt that accurately describes the bird but uses features shared across classes.

TODO 2.2.2 - Worst Performing Prompting Strategies

The hardest classes to distinguish were the sparrows, Henslow's Sparrow and Tree Sparrow, because their prompts necessarily overlap in vocabulary: both are small, both are brown/streaky, and both are ground-dwelling. Even accurate descriptions like "a tiny streaky sparrow with an olive-colored head" can confuse CLIP because the word "sparrow" activates a shared visual prototype, and subtle differences in head color (olive vs. rufous) may not be salient enough in CLIP's embedding space to separate them reliably.

Prompts that rely on behavioral or habitat cues not visible in a static image also perform poorly. Describing the Palm Warbler as having "a habit of bobbing its tail" is accurate ornithologically but invisible in a photograph. CLIP cannot infer behavior from a still image, so this portion of the prompt adds noise rather than signal. The lesson is that CLIP matches text to visible image content; any prompt feature that cannot be directly seen in the image does not help and may hurt by shifting the embedding away from the visual features that do discriminate.

Generic structure like "a photo of a [description]" adds no benefit over a bare description for fine-grained classification, since all classes share the same photographic context. Prompt templates designed for coarse-grained zero-shot tasks (like ImageNet) do not transfer well to fine-grained bird classification where inter-class visual similarity is high.